VICTOR IBHAFIDON

Graduate Technical Advisor

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PROFESSIONAL SUMMARY

My approach to software engineering is grounded in the belief that good software is built through thoughtful design, sound engineering, and continuous iteration. I've worked on AI/ML solutions across software engineering, data, security, and cloud, gaining experience through building, iterating, and learning from challenges along the way, while balancing technical requirements with broader business needs. My philosophy is simple: I don't just want something that works in a demo. I want to understand what can go wrong, prove it works, and build it properly. I have trained, evaluated and deployed computer-vision models, including PyTorch CNNs served through production APIs.

TECHNICAL SKILLS

Backend & Data: FastAPI, Django, REST APIs, WebSockets, Microservices, PostgreSQL, MySQL, MongoDB, Redis, Neo4j, pgvector, Vector Search, Data Pipelines, ETL

Programming: Python, TypeScript, JavaScript, SQL, Bash

Cloud & DevOps: AWS (EC2, S3, Lambda, IAM, EKS), Microsoft Azure, Azure OpenAI, Docker, Kubernetes, GitHub Actions, CI/CD, Linux, Nginx, OpenTelemetry, Prometheus, Grafana

AI & Machine Learning: LLMs, Generative AI, AI Agents, Multi-Agent Systems, LangGraph, MCP, JSON-RPC, Tool Calling, RAG, GraphRAG, Prompt Engineering, AI Evaluation, PyTorch, Transformers, Scikit-learn, Computer Vision, NLP

Security & AI Governance: AI Security, AI Governance, Authentication, Authorization, RBAC, API Security, OWASP Top 10, Prompt Injection Mitigation, AI Safety, Responsible AI, GDPR

MLOps & Model Operations: MLflow, Model Training, Model Evaluation, Model Monitoring, Drift Detection, Experiment Tracking, Model Deployment

PROFESSIONAL EXPERIENCE

Software Engineer (AI/ML), Bastion Shield Technologies

London, United Kingdom | April 2026 – Present

- Built Bulwark, a multi-tenant AI agent governance platform that gives organisations control and visibility over AI agents. Designed and implemented agent identity and lifecycle management, policy enforcement, human oversight workflows, and detection of unregistered agents, orphaned ownership, and stale credentials.
- Engineered the backend in Python, TypeScript, and PostgreSQL, implementing RBAC, server-side tenant isolation, secure authentication, API-key management, and an SDK for external teams to register agents and stream policy decisions. Designed tamper-evident audit logging with offline verification, strengthened data privacy controls, and remediated four production vulnerabilities, including a session-token forgery path.
- Owned the cloud infrastructure using Terraform-managed AWS environments across development, staging, and production. Containerised the platform with Docker and delivered it through GitHub Actions CI/CD, while working on distributed system patterns including queues, event streaming, caching, and horizontal scaling to improve performance and latency.
- Developed behavioural anomaly detection combining statistical methods with usage forecasting to identify unusual agent activity. The system achieved a 0.981 ROC-AUC against an Isolation Forest baseline and was integrated into the TypeScript runtime for real-time scoring.

Full Stack Software Engineer, Basketball Nxtion

London, United Kingdom | February 2026 – April 2026

- Developed the MVP of Plxyground, a mobile application connecting basketball players, creators, and sports communities, contributing across product design, application architecture, backend development, APIs, authentication, and user workflows.
- Designed the PostgreSQL data architecture and analytics pipelines for the MVP, structuring user, activity, engagement, and platform data to support data ingestion, transformation, analysis, and business intelligence. Built data workflows enabling insights into platform performance, user behaviour, engagement, and product adoption.
- Designed and integrated an AI architecture connecting LLM/ML services with PostgreSQL data, backend APIs, and application business logic. Built workflows for personalised recommendations, intelligent user matching, and content discovery, using user and platform data to generate relevant results, improve engagement, and automate discovery workflows.
- Deployed and maintained the MVP infrastructure on AWS, implementing Docker, CI/CD, testing, rollbacks, monitoring, and third-party integrations while collaborating with frontend engineers and stakeholders throughout the software development lifecycle from requirements and development through testing, deployment, and iteration.

Full Stack Python Engineer Intern, CyberAI Technologies

London, United Kingdom | September 2025 – November 2025

- Collaborated with senior AI engineers to develop Surfyn AI, a conversational AI customer support platform enabling businesses to deploy knowledge-based AI assistants grounded in websites and documents to automate customer interactions and provide relevant, context-aware responses.
- Contributed to the development of RAG and conversational AI workflows using Python and Django, working on document ingestion, chunking, embeddings, vector-based retrieval, prompt construction, conversation context, REST APIs, and LLM integrations while learning how knowledge-based AI systems are designed and evaluated.
- Developed backend functionality and AI workflow improvements, working on API development, debugging, testing, performance optimisation, knowledge retrieval, and integration between AI services and the Django application.
- Worked with the AI Security Engineering team on AI evaluation, testing, safety, and deployment workflows, helping investigate retrieval quality, response relevance, hallucination risks, prompt injection, sensitive data handling, logging, and application reliability across production AI systems.

SELECTED PROJECTS

Portivo Estate Suite (AI Powered PropTech Operating System): Approached Portivo from a consulting and product-engineering perspective, working with estate managers to understand how AI could be integrated into existing property operations to reduce administrative workload and improve lettings, sales, maintenance, compliance and reporting. Worked directly with users through discovery, workflow design, testing and implementation, translating feedback into product requirements, documenting APIs and operational processes, validating end-to-end workflows and supporting customer configuration and adoption. Designed the system architecture and took ownership of end-to-end development across product analysis, UX/UI, frontend, backend, AI/ML, data engineering, cloud, security, QA, AI governance and integrations. Built the multi-tenant application with React, TypeScript, Next.js and Tailwind, developing dashboards, property management, CRM, reporting and administrative workflows, backed by Python, FastAPI and SQLAlchemy for business logic and APIs. Designed the PostgreSQL data model with tenant-scoped queries and Alembic migrations against a live customer database, using Redis and Celery for caching, asynchronous processing and scheduled jobs. Built the AI architecture using LangGraph, Anthropic Models, RAG, OCR, chunking, embeddings and pgvector, with an orchestrator delegating typed tasks to specialist agents for document retrieval, lease abstraction, arrears detection, compliance tracking, maintenance triage and outbound sales prospecting. Implemented MCP servers and scoped credentials to give agents controlled access to CRM, email and document tools rather than creating bespoke integrations. Evaluated AI performance using labelled datasets for groundedness and extraction accuracy, with Langfuse for tracing, schema-validated outputs, confidence thresholds and audit trails for agent actions. Applied RBAC, tenant isolation, scoped permissions, security controls and AI governance to protect customer data and control agent behaviour, including prompt-injection and data-leakage risks. Containerised and deployed the platform to the cloud using Docker and GitHub Actions CI/CD, while designing API integrations, technical documentation and implementation workflows to support enterprise adoption. **Live in production at:** portivoestatesuite.com

Autonomous AI Defence System for Critical Infrastructure: Designed an autonomous surveillance and anomaly-detection system for critical infrastructure such as rail corridors, electrical substations, solar and wind farms, ports, reservoirs and industrial sites. Drones and ground vehicles can run scheduled or operator-triggered missions in simulation, using computer vision to detect intrusions, trespassing, vegetation encroachment and equipment anomalies. Designed the architecture as an event-driven system where simulated vehicles generate telemetry and camera data, computer vision services process observations, and Kafka distributes events to the backend for alerting, tracking and decision-making. Built FastAPI services and PostgreSQL for mission, asset and operational state, Redis for fast-access state and coordination, and Next.js/WebSockets for a live operations dashboard showing vehicle locations, telemetry, alerts and mission status. Designed a bounded autonomy layer that evaluates detected events against predefined rules and operational constraints before proposing actions, with high-impact actions routed to a human operator for approval, override or emergency stop. Integrated YOLO/PyTorch for computer vision inference and designed a hardware integration layer using MQTT and ROS to connect mission and control services with drones and ground vehicles through standardised telemetry, command and sensor interfaces, separating vehicle hardware from the core application and allowing the same architecture to support simulation and physical robotics hardware. Added an audit layer that records missions, detections, decisions, approvals and commands so actions can be reconstructed and reviewed. Designed the ML lifecycle around model training, evaluation, fine-tuning, drift detection and retraining triggers, with Prometheus/Grafana for system and model observability. Containerised the services with Docker and designed Kubernetes, Terraform and AWS infrastructure for repeatable deployment, while implementing JWT, RBAC, OWASP-aligned protections, input validation, SSRF/injection safeguards and explicit failure-mode handling. PyTorch CNN models were trained and evaluated across the 231,444-image dataset to up to 92% classification accuracy, then served as a versioned inference service through Docker and REST APIs. GitHub: github.com/Web8080/Autonomous_AI_defense_system_design

EDUCATION

MSc Data Science, Nottingham Trent University, United Kingdom (2023 – 2024)

BSc Computer Science, Ambrose Alli University, Nigeria (2016 – 2021)